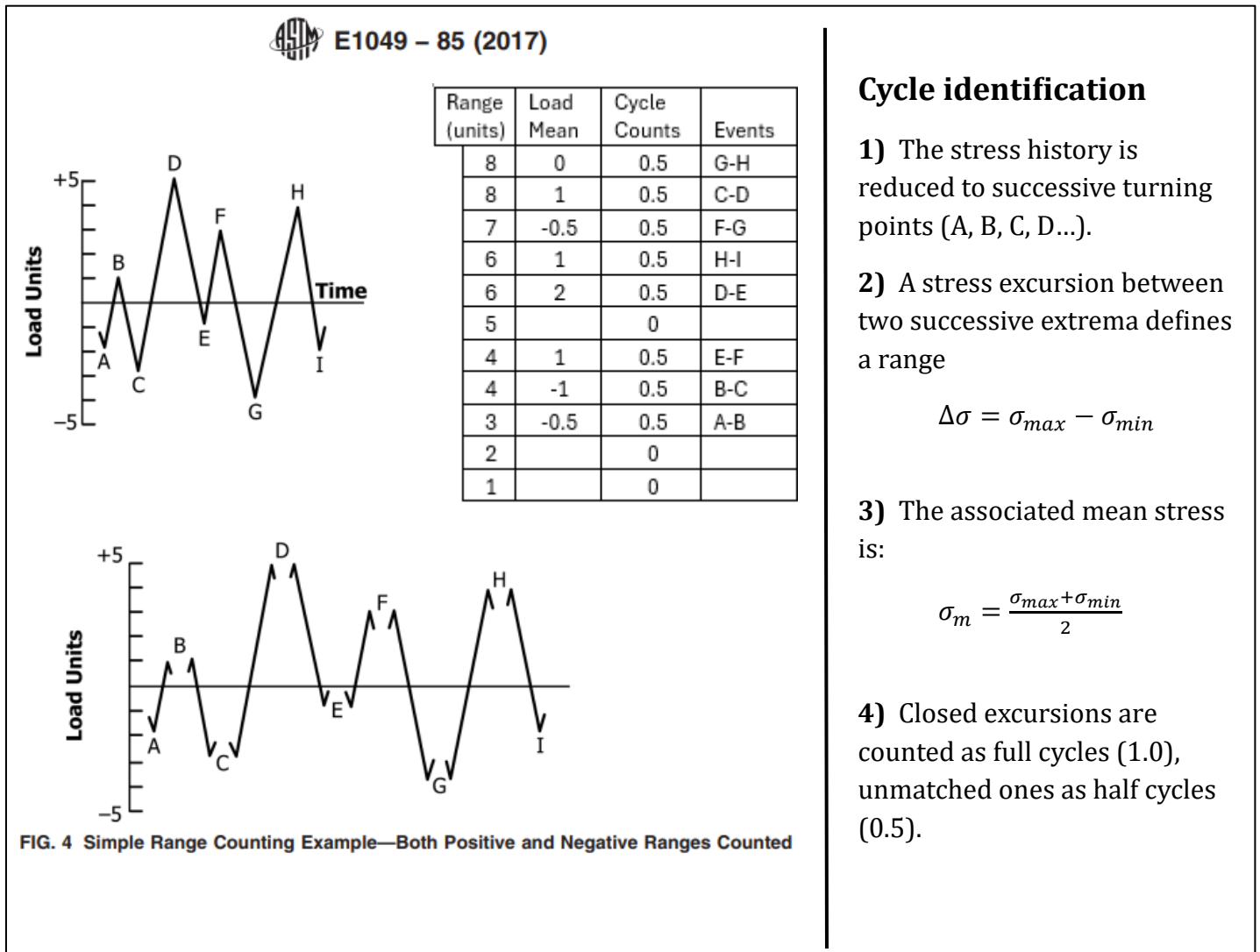


Range-Mean Method

Principle

The range-mean method counts stress excursions between successive extrema while preserving the associated mean stress of each cycle. Each counted cycle is therefore defined by its stress range and its mean value.

ASTM example



Assumptions & Recommended use

Fatigue damage **depends on both stress amplitude and mean stress.**

By preserving the mean value of each cycle, the range-mean method allows **direct integration with Constant Life Diagrams and mean stress correction models.**

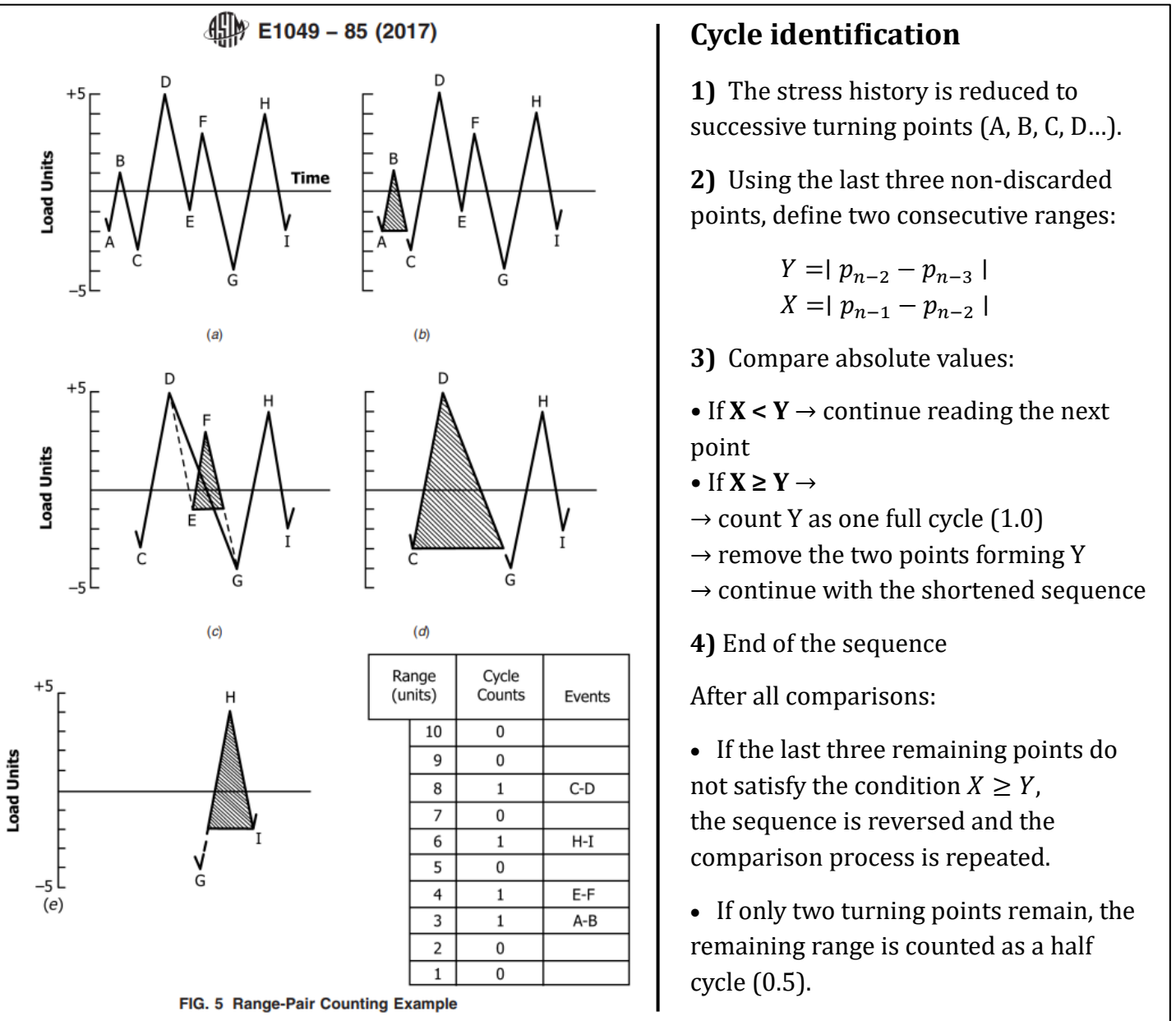
Range-Pair method

Principle

The range-pair method identifies closed stress cycles by comparing adjacent ranges and eliminating smaller inner excursions.

A cycle is counted when a stress range is enclosed by a later range of equal or greater magnitude.

ASTM example



Assumptions & Recommended use

The method assumes that **fatigue damage is governed by closed stress cycles**.

For load histories with many reversals (e.g., aircraft spectra), the influence of isolated half cycles becomes negligible, making this assumption physically consistent.

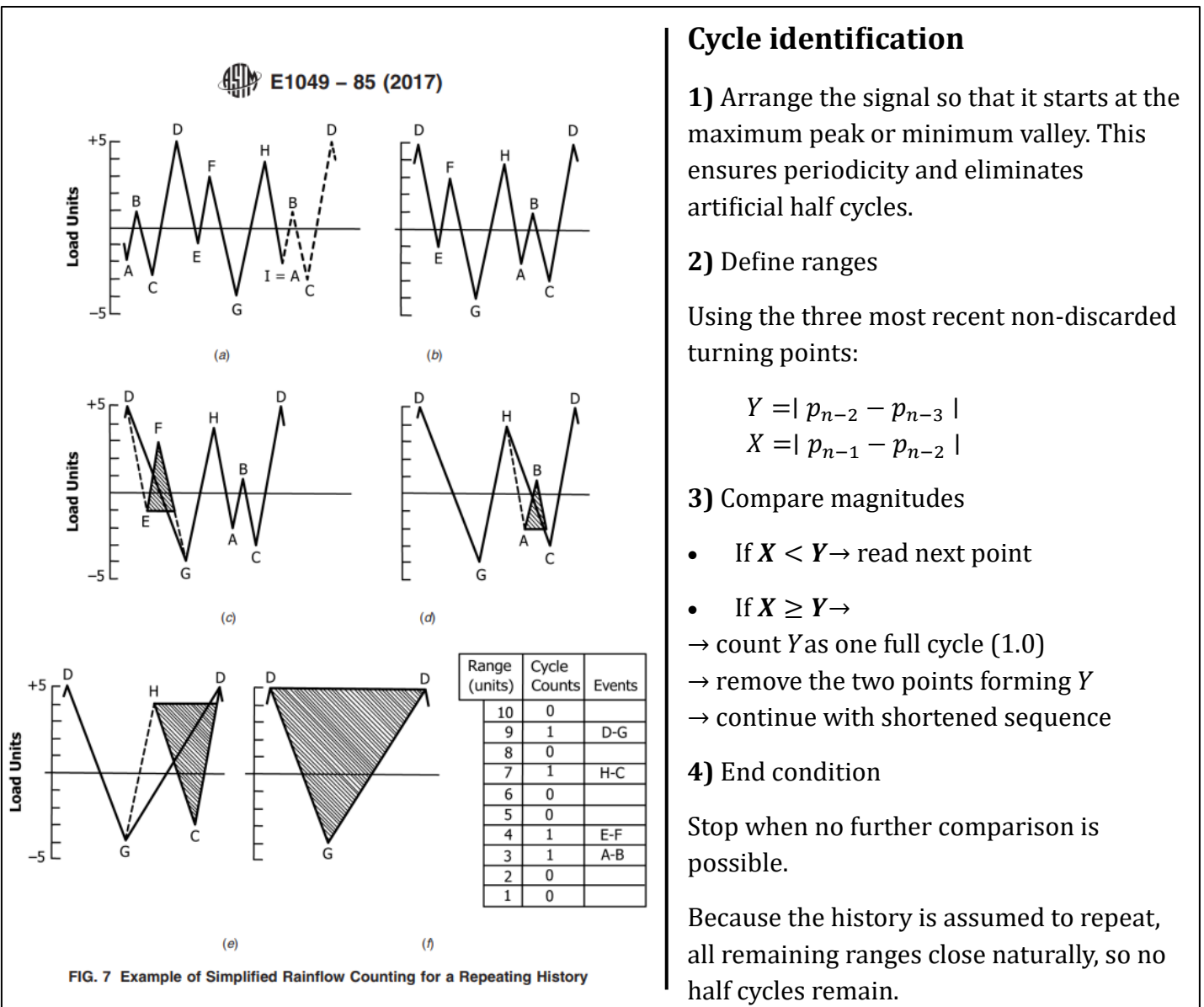
Simplified Rain Flow method

Principle

The simplified rainflow method is a variant of rainflow counting intended for repeating load histories. The signal is rearranged so that it starts at the global maximum peak or minimum valley. Cycles are then extracted by comparing adjacent ranges and counting only closed full cycles.

For a repeating history, this method produces only full cycles (no half cycles).

ASTM example



Assumptions & Recommended use

The method assumes that the load history is cyclically repeated.

Under this assumption, fatigue damage is governed by closed stress-strain hysteresis loops, and only full cycles are relevant.

It provides a consistent cycle count for **periodic spectra without residual half cycles.**

Sources

- ASTM E1049 – 85 (2017), *Standard Practices for Cycle Counting in Fatigue Analysis*